

September 21, 2026

Dear Rachel Ko,

I am writing to apply for the Software Engineer, GKOS position at Magnet Forensics. With a Master of Science in Computer Science from the University of Calgary and hands-on experience building Python and C++ software, backend systems, data pipelines, AI workflows, and Linux-based research software, I am excited by the opportunity to contribute to Graykey platform work that helps investigative teams build safer, faster, and more consistent user experiences.

What stands out to me about Magnet is that the software has a real public-safety mission behind it. I like engineering work where quality, reliability, usability, and careful technical decisions matter because the users are doing important work. The GKOS role's focus on building features, addressing bugs and technical debt, collaborating with UX, and supporting automated UI testing fits the kind of practical, quality-oriented software engineering I want to keep growing in.

My graduate research focused on building reliable software and data systems for large Earth observation datasets. I developed C++ and Python pipelines for processing, encoding, storing, and retrieving satellite imagery; automated preprocessing workflows; and improved one encoding workflow from 233 seconds to 26 seconds through multithreaded implementation. This work required Linux-based development, Git, debugging, documentation, attention to edge cases, and performance-minded problem solving.

I also bring backend and product engineering experience from my work at Asr Gooyesh Pardaz, where I developed Django REST APIs backed by PostgreSQL for NLP, text-to-speech, and speech-to-text products. I built authentication workflows, implemented payment-service logic, and contributed to a WebRTC-based chatbot system using GPT-style responses and FAQ data. These experiences helped me develop the habit of writing maintainable code, investigating failures carefully, and connecting technical work to user-facing outcomes.

The role's AI/ML nice-to-have is also a strong match for my background. I have built PyTorch training and evaluation workflows, worked with model performance and data quality, and I am building an agentic automation platform using LLM workflows, Docker-based services, Google Sheets, Google Drive, Telegram API, and workflow automation tools. I would be ready to deepen my practical exposure to React, hardware-device workflows, networking fundamentals, and digital-forensics products, while contributing immediately from my Python, Linux, Git, Docker, CI, AI/ML, debugging, and cross-team collaboration strengths.

Thank you for considering my application. I would welcome the opportunity to bring my software engineering, AI workflow, data-systems, and research software background to Magnet Forensics and the GKOS team.

Sincerely,
Amir Mirzai Golpayegani
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